The empirical recurrence for OEIS A268808, recovered from its tail

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Abstract

OEIS A268808 records "Empirical recurrence of order 68" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 692$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{68} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A268808 is "Number of nX7 0..2 arrays with some element plus some horizontally, vertically or antidiagonally adjacent neighbor totalling two not more than once.". It has offset 1 and begins

$$768, 4914, 48460, 640926, 8734264, 115671422, 1508087180, 19390335102, \ \ldots$$

The entry states:

> Empirical recurrence of order 68 (see link above)

As of the "Last modified" line on the live entry (Feb 13 2016 18:25:53, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 692$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 692$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1384$ exact terms from $n = 4$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{68} c_i\, a(n-i),$$

c_1	2	c_{18}	−40352987546	c_{35}	−2516449266318	c_{52}	392403435
c_2	169	c_{19}	−1731681754	c_{36}	−5299006856467	c_{53}	121804600
c_3	694	c_{20}	169766303246	c_{37}	4612288755692	c_{54}	−56942553
c_4	−7705	c_{21}	−172083944226	c_{38}	−370411564193	c_{55}	−45190096
c_5	−81246	c_{22}	−432283697779	c_{39}	−1670643254910	c_{56}	−2736961
c_6	−222034	c_{23}	1174345160584	c_{40}	881083365721	c_{57}	8047064
c_7	517502	c_{24}	−213727795318	c_{41}	254923161724	c_{58}	3525933
c_8	4048112	c_{25}	−3410566119732	c_{42}	−375390897947	c_{59}	53154
c_9	1950100	c_{26}	6248797331497	c_{43}	18817875590	c_{60}	−513311
c_{10}	−34597333	c_{27}	−1297199940236	c_{44}	107541892314	c_{61}	−229598
c_{11}	−49390138	c_{28}	−14252531123312	c_{45}	−21340705994	c_{62}	−42389
c_{12}	222391082	c_{29}	31972773053846	c_{46}	−26802863398	c_{63}	1728
c_{13}	411980164	c_{30}	−36845671545670	c_{47}	6427885660	c_{64}	2624
c_{14}	−1316829914	c_{31}	22145471845712	c_{48}	6733400845	c_{65}	486
c_{15}	−2225642966	c_{32}	1754955549210	c_{49}	−1016537708	c_{66}	3
c_{16}	7627596454	c_{33}	−16611707402840	c_{50}	−1726204662	c_{67}	−10
c_{17}	7520225330	c_{34}	14194215422343	c_{51}	−78736390	c_{68}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A268808 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{68-1} - \cdots - c_{68}$. Its constant term is $-c_{68} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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